[FORENSIC PSYCHOLINGUISTIC & DECEPTION DETECTION ARCHITECTURE]

SUBTEXT ENGINE

An algorithmic forensic engine for detecting covert manipulation, systemic deception, reflexive control, and testimony fabrication across high-stakes negotiations, private equity due diligence, and elite corporate litigation.

21

Peer-Reviewed Scientific Treatises (FBI, BGH 1999, Cambridge, LIWC, RAS)

8

Global Forensic Disciplines (Statement Analysis, CBCA, Reflexive Control)

0%

Data Leakage: Air-Gapped Private VPC & On-Premise Local Deployment

100%

Court-Admissible Metrics: Strictly excludes discredited commercial SCAN

The Dangerous Blind Spot of Generative AI

Why standard LLMs and human analysts fail to detect sophisticated deception and bad-faith actors

⚠️

1. The Content-Layer Vulnerability

Standard LLMs (GPT-4o, Claude 3.5, Gemini) and traditional legal counsel evaluate communication at the **semantic content layer** (*what* is said). Skilled deceivers, predatory partners, and hostile state actors exploit this by authoring syntactically pristine narratives that conceal fraudulent intent beneath immaculate rhetoric.

🤖

2. RLHF-Induced Sycophancy

Reinforcement Learning from Human Feedback (RLHF) forces public AI models to be polite, agreeable, and trustful. When presented with a manipulative contract or evasive executive testimony, standard LLMs default to "charitable interpretation," actively rationalizing glaring temporal gaps and psychological red flags.

🔬

3. The Subtext Engine Solution

Subtext Engine operates exclusively at the **structural, syntactic, and reflexive layers**: *how* things are articulated, *what* critical experiential details are omitted, and *how* the listener's cognitive filter is being engineered. We replace superficial sentiment analysis with verifiable forensic psycholinguistics.

METHODOLOGICAL FOUNDATION // 21 SCIENTIFIC TREATISES

STANDARDS: **BGH 1999 // DAUBERT // FRYE**

The 8 Global Forensic Intelligence Pillars

A mathematically unified framework validated by German, European, and US judicial precedents

PILLAR 01 // STATEMENT ANALYSIS

Statement Balance & Syntax Stalling

FBI 20/50/30 structural balance, 1st-person pronoun drop, reliable denials ("I did not [X]"), and temporal lacunae (Adams, McClish, Coulthard, Olsson).

PILLAR 02 // JUDICIAL CREDIBILITY

CBCA & Reality Monitoring

19 Undeutsch criteria of experiential memory (German Federal Court of Justice BGH 1999 standard). Cognitive load elevation in deceit (Steller, Köhnken, Vrij).

PILLAR 03 // COMPUTATIONAL METRICS

Quantitative LIWC & LDI Ratios

Mathematical function word ratios, exclusive particle dropping, and the Linguistic Distancing Index (LDI) for psychological detachment (Pennebaker, Newman).

PILLAR 04 // REFLEXIVE CYBERNETICS

Reflexive Control & Coercion

Lefebvre's reflexive polynomials, false double-binds, Hassan's B.I.T.E. model, and Biderman's UN interrogation manipulation matrices.

Additional Modules: Cognitive Alethiology (Sartori aIAT), Affective Mimicry & Simulated Remorse (Porter), Global Evaluation System (Arce SEG), and Cross-Cultural SUE Calibration (Nortje). 21 FOUNDATIONAL TREATISES

Legal Guarantee: Zero unscientific commercial SCAN gimmicks; 100% peer-reviewed academic literature

ENTERPRISE APPLICATIONS // OPERATIONAL THEATERS

DEPLOYMENT CONTEXT: **HIGH-VALUE TRANSACTIONS**

Four High-Stakes Commercial Arenas

Where subtle deception, narrative inflation, and hidden agendas cost millions of dollars

1. Private Equity & M&A Due Diligence

INVESTMENT & M&A

- **Founder & Management Authenticity Audits:** Screening executive interviews, pitch transcripts, and confidential data room memoranda for unacknowledged liabilities.
- **Narrative Inflation Detection:** Pinpointing linguistic markers of exaggerated ARR retention, customer churn rationalization, and IP ownership ambiguity.
- **Shadow Negotiation Profiling:** Uncovering counterparty BATNA, hidden friction points, and deal-breaking redlines before signing Definitive Agreements.

2. Corporate Investigations & Forensic Accounting

FORENSIC RISK & GRC

- **Whistleblower Credibility Scoring:** Filtering genuine multi-million-dollar corruption/bribery tips from vexatious or fabricated competitor smears using CBCA.
- **Procurement Fraud Collusion:** Detecting terminology mutation across ostensibly independent supplier tender dossiers.
- **Internal Fraud Interviews:** Assisting forensic accountants and corporate investigators in identifying rehearsed alibis and deceptive evasion.

3. Elite White-Collar Defense & Cross-Border Arbitration

BIG LAW & ARBITRATION

- **Deposition Preparation & Cross-Examination:** Stress-testing key witness statements under Coulthard & Olsson forensic linguistic models before depositions.
- **International Arbitration (LCIA, ICC, SIAC, ICSID):** Auditing written witness statements for institutional coaching, ghost-drafting contamination, and syntax stall.
- **Expert Witness Submissions:** Providing objective, methodology-backed forensic psycholinguistic reports admissible under Daubert and European standards.

4. Enterprise Security & Counter-Reflexive Intelligence

GOVTECH & DEFENSE

- **Insider Threat Early Warning:** Monitoring communicative anomalies, linguistic distancing shifts, and radicalization/coercion markers in sensitive internal communications.
- **Countering Reflexive Manipulation:** Shielding corporate boards and negotiating teams from adversarial state/competitor influence campaigns (Lefebvre matrices).
- **Hostile Negotiation Shield:** Real-time linguistic telemetry during high-tension hostage/extortion crises or regulatory investigations.

Strategic Impact: Replacing subjective gut instinct with mathematically formalized forensic science

MARKET LANDSCAPE // GLOBAL ENTERPRISE MATRIX

TOTAL ADDRESSABLE MARKET: **$12.4B ENTERPRISE GRC & LITIGATION**

Top 20 Global Target Institutions

Decision-makers, critical pain points, and specific Subtext Engine integration vectors

INSTITUTION / GLOBAL TIER	TARGET DECISION-MAKER (BUYER)	CRITICAL VULNERABILITY & COST OF FAILURE	SUBTEXT ENGINE SOLUTION VECTOR
Kroll / FTI Consulting / Control Risks / Alvarez & Marsal `Forensic Investigations`	Global Heads of Investigations, Forensic Risk Managing Directors, Senior Partners	Rehearsed alibis, hidden asset transfers, subjective investigator bias, unverified whistleblower tips costing clients $50M+.	Automated CBCA / Statement Analysis scoring of interview transcripts and confidential informant dossiers.
Kirkland & Ellis / Latham & Watkins / Skadden / Quinn Emanuel `Big Law White-Collar Defense`	Chairs of White-Collar Defense, Senior Litigation Partners, International Arbitration Heads	Coached witness testimony unraveling on the stand; unprovable concealed intent in complex civil fraud or SEC enforcement.	Deposition stress-testing; institutional syntax contamination audit; forensic linguistic expert evidence reports.
Blackstone / KKR / Carlyle Group / Insight Partners `Private Equity & Growth Capital`	Managing Directors, Operating Partners, Due Diligence Heads, Investment Committee Members	Investing $200M+ based on fabricated metrics, inflated customer satisfaction narratives, or concealed founder IP disputes.	Linguistic Distancing Index (LDI) and deceptive omission audits on founder management presentations and Q&A transcripts.
Palantir Technologies / Booz Allen Hamilton / Dataminr `Defense & Intelligence Contractors`	Chief Risk Officers, Threat Intelligence Directors, Heads of Counter-Intelligence Practices	Adversarial reflexive control operations; cognitive deception campaigns by state-backed actors; malicious insider threats.	Integration of Lefebvre reflexive polynomial algorithms into enterprise threat intelligence pipelines.
McKinsey & Co. / Bain & Co. (PE Practice) / BCG `Strategic Management Advisory`	Senior Partners (Restructuring & M&A), Heads of Strategic Risk, Board Advisory Leads	Advising Fortune 500 boards on cross-border joint ventures while blind to counterparty bad-faith negotiating tactics.	Shadow negotiation profiling; real-time cognitive load mapping during cross-border M&A negotiations.

Combined Annual Transaction Volume: $1.8 Trillion+ | High ROI: Eliminating a single catastrophic M&A error

FORENSIC CASE STUDY // LIVE DECONSTRUCTION

CASE DISCIPLINE: **DECEPTION DETECTION**

Anatomy of Deception: A $120M M&A Deposition

Contrasting human surface review against the Subtext Engine structural X-Ray

RAW FOUNDER TESTIMONY (M&A LITIGATION)

> *"I was actively leading our enterprise expansion in Q3. We diligently executed all tier-one customer onboarding protocols. There were no material cancellations reported to me personally. Around mid-August, the platform upgrade was deployed. Everything was operating according to roadmap specifications. A decision was made to postpone enterprise reporting due to ongoing architectural scaling. I would never misrepresent our core churn metrics to incoming private equity investors."*

Typical Corporate Due Diligence Assessment:
"Founder exudes confidence, accepts leadership role, explicitly denies customer cancellations, and defends company integrity."

SUBTEXT ENGINE FORENSIC VERDICT

🔴 **1. Pronoun Evaporation & Responsibility Severing:** Moves from personal agency (*"I was leading"*) to passive evasion (*"A decision was made to postpone"*). In Statement Analysis, sudden agent deletion at the precise point of failure marks institutional guilt.

🔴 **2. Equivocal Subjunctive Denial:** *"I would never misrepresent..."* is a hypothetical non-denial, NOT a reliable factual denial (*"I did not misrepresent"*). McClish Statement Law: innocent subjects make direct factual assertions in the simple past tense.

🔴 **3. Subjective Memory Void (CBCA Criteria 3 & 4):** Zero perceptual, spontaneous, or sensory details regarding the mid-August upgrade; high cognitive friction around customer retention numbers.

⚠️ **Litigation Strategy Recommendation:** Subpoena internal Slack communications between Aug 10–25 regarding churn alerts. Founder was fully aware of catastrophic enterprise customer cancellations prior to closing.

Outcome: The investment committee avoided a $120M write-down after confirming 40% enterprise contract churn

COMMERCIAL ENGAGEMENT // DEPLOYMENT ARCHITECTURE

INTEGRATION TIERS: **ENTERPRISE SLA // SOC2 / HIPAA COMPLIANT**

Enterprise Deployment & Partnership Models

From bespoke high-stakes shadow advisory to enterprise air-gapped on-premise integration



TIER 01

Forensic Advisory & Shadow Desk

$25K – $75K / deal

Bespoke, rapid-response forensic profiling for active M&A transactions, hostile takeovers, or emergency whistleblower crises under strict attorney-client privilege.

- 24–48 hour turnaround on dossiers
- Counterparty shadow-profiling
- Pre-deposition testimony auditing



TIER 02 (FLAGSHIP)

Air-Gapped Private VPC Deployment

$120K – $350K / yr

Full enterprise integration within your isolated corporate cloud (AWS GovCloud / Azure Dedicated / On-Premise bare metal). Zero external API calls, zero telemetry leakage.

- Local weights (Llama 3 / Mistral / DeepSeek)
- Batch processing of 100K+ legal discovery docs
- Enterprise SLA, custom API & prompt fine-tuning



TIER 03

OEM & LegalTech Platform License

Custom Royalty

Embed Subtext Engine's 21 algorithms directly into your proprietary GRC platform, e-Discovery software, or ATS HR background verification ecosystem.

- White-label API & SDK libraries
- Full access to raw psycholinguistic matrices
- Co-branded academic validation credentials

Return on Investment: Preventing a single toxic executive hire or fraudulent acquisition delivers infinite ROI

Initiate a Confidential Proof of Concept

Evaluate Subtext Engine in a blinded trial using your organization's historical resolved cases

The Blinded Proof of Concept (PoC) Protocol

Provide us with 3 anonymized, historical investigative texts (e.g., fraud depositions, contested whistleblower reports, or negotiation transcripts where the true factual outcome is already known to your team).

Our engine will generate a blinded forensic teardown report. Compare our structural psycholinguistic conclusions against ground truth to confirm the engine's diagnostic fidelity firsthand.

```
$ agy install elevchuk-lab/subtext-engine
```

Laboratory Credentials & Executive Contact

👤 **Yevgeniy Levchuk** (Principal Investigator, elevchuk-lab)

🏛️ **Research Focus:** Computational Cognitive Forensics & Algorithmic Deception Detection

🔗 **Open Source Core:** github.com/elevchuk-lab/subtext-engine

📚 **Academic Registry:** ORCID Verified // CERN Zenodo Research Archive

⚖️ **Licensing:** MIT Open Source Core / Enterprise Commercial Dual-License

📩 **Inquiries:** Via GitHub Institutional Channels & Confidential Retainers

* The core engine is fully open-source for academic peer-review, security vetting, and algorithmic transparency prior to enterprise deployment.